

Intro to Quantum Computing - Week 1 Notes

1 Quantum Postulates

- Postulate one: How to model states: Every closed quantum state can be represented by a vector $|\psi\rangle$ in a vector space in $\mathbb{C}^n$ ($\mathbb{R}^n$), this can be $|\psi\rangle = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$, where the vectors are normalized, meaning that $|c_1|^2 + |c_2|^2 + \dots + |c_n|^2 = 1$. If we have a collection of vectors which are part of the set of V , that is $|r\rangle + |w\rangle \in V$, for $\alpha |v\rangle \in V$
- Postulate two: How can we measure: Measurements are described by a set of operators that act on the state vector.
- Postulate three: State evolution: The time evolution of a closed quantum system is governed by the Schrödinger equation.
- Postulate four: Composite systems: The state of a composite quantum system is the tensor product of the states of its components.

Closed systems are deterministic